

Question 1:

A project requires an initial investment of ₹100,000 and is expected to generate additional revenues of ₹30,000 per year with additional costs of ₹10,000 per year for 5 years. Assume an annual inflation rate of 3%. Calculate the Incremental Cash Flows, adjusted for inflation.

Question 2:

A company is evaluating a project with an initial investment of ₹60,000. The project is expected to generate revenues of ₹15,000 in the first year and grow by 4% annually due to inflation. The costs are ₹5,000 in the first year and grow by 2% annually due to inflation. The project lasts for 4 years. Calculate the Incremental Cash Flows, adjusted for inflation.

Question 3:

Calculate the NPV for a project with the following cash flows, considering a discount rate of 10%:

- Initial investment: \$100,000
- Year 1: \$30,000
- Year 2: \$40,000
- Year 3: \$50,000

Question 4:

A company is considering a new project that requires an initial investment of \$50,000. The project is expected to generate additional revenues of \$30,000 per year and additional costs of \$5,000 per year for the next 4 years. Calculate the Net Present Value for the project, considering a discount rate of 10%.

Question 5:

A company is evaluating a project with an initial investment of \$60,000. The project is expected to generate revenues of \$35,000 and operating expenses of \$5,000 annually. The

inflation rate is 5%. The project lasts for 4 years. Calculate the NPV, considering a discount rate of 10%.

Question 6:

Calculate the NPV considering:

- Initial investment: \$150,000
- Year 1-4 cash inflows: \$45,000 annually
- Working capital requirement: 10% of annual inflows
- Inflation rate: 5%
- Discount rate: 12%

Question 7:

A company plans to invest \$5 million in new manufacturing equipment. The equipment is expected to increase revenues by \$2.5 million annually for 5 years. The annual maintenance cost is \$0.5 million, and wages will increase by \$0.3 million per year. Additional working capital of 12% of the annual revenue increase is required at the start of each year. Working capital will be released at the end of the project.

The inflation rates are as follows:

- Revenues: 8%
- Wages: 4%
- Maintenance: 5%
- General prices: 6.5%

The company's real cost of capital is 8%.

Calculate the net annual cash flows of the project and determine NPV.

Question 8:

A company invests \$100,000 in equipment with a useful life of 4 years and salvage value of \$5000. It will be depreciable on WDV basis at 30%. The project is expected to generate

additional revenues of \$40,000 per year and additional costs of \$10,000 per year. The corporate tax rate is 30%. Ignore inflation. Calculate the NPV, considering a discount rate of 10%.

Question 9:

An inventory management system will cost Rs.1crore. It will be depreciable on SLN basis for 5 years which is its useful life. At the end of 5th year, it can be disposed for Rs.20 lakhs. This system will annually save Rs.30 lakhs before depreciation and tax. The tax rate is 30% and the system will require Rs. 10 lakhs initially as working capital. Assuming 10% cost of capital, calculate the net investment, net annual cash flows, terminal cashflow & NPV of this project.

Question 10:

XYZ Ltd is considering introduction of new product the firm estimates that it can sell annually 5000 units of a product at Rs. 20/unit. The cash variable expenses to manufacture and sell the product are estimated at Rs. 12/unit. It will also involve fixed cost of Rs. 5000/annum. The Plant to manufacture the product is available for Rs.100,000. further Rs. 20,000 will be needed for installation of the machinery. The salvage value of the plant after its life of 10 years is estimated to be Rs.5000. The working capital investment of Rs.30,000 would be required in the year during the installation of the plant. The firm uses Straight line method of depreciation on original cost of the assets, ignoring salvage value of fixed asset. Assuming the tax rate to be 50%, you are required to calculate, initial investment, ACF And TCF.

Question 11:

Avighna Technologies, a mid-sized industrial robotics firm, is evaluating a proposal to launch a new line of automated material-handling equipment. The project has a life of 4 years and requires an initial investment in machinery costing ₹40,00,000. The machinery will be depreciated using the straight-line method over 4 years, with no salvage value. The company expects to sell 1,000 units annually at a price of ₹12,000 per unit in the first year, with selling prices expected to grow at a general inflation rate of 4% per annum. The variable cost per unit is ₹6,000, expected to grow at a specific inflation rate of 6% per annum due to input cost pressures. Fixed operating costs (excluding depreciation) are ₹10,00,000 in Year 1 and are also expected to increase by general inflation of 4% annually.

To support operations, the project will require net working capital investment equal to 10% of the projected sales revenue each year. The working capital is to be invested at the beginning of each year and will be fully recovered at the end of the fourth year. The company's tax rate is 30%, and the project's nominal required rate of return is 14% (including general inflation). All cash flows occur at year-end unless otherwise stated. Calculate the Net Present Value (NPV) of the project. Should Avighna Technologies undertake the project?

Question 12:

EcoGreen Technologies, a company focused on sustainable innovations, is considering an initial investment of \$2,500 in a new project. The project is expected to generate cash inflows of \$600, \$600, \$700, \$700, and \$820 at 12-month intervals. The project does not have any residual value. The required rate of return for financing the project is 9%. Calculate the IRR.

Question 13:

A project with the following cash flows is under consideration:

- Initial investment: \$2.5 million
- Year 1: 1.1 million
- Year 2: 860,000
- Year 3: 700,000
- Year 4: 720,000
- Year 5: 820,000

Assuming 9% cost of capital, calculate the Discounted Payback Period.

Question 14:

A project with the following cash flows is under consideration:

- Initial investment: \$2.5 million
- Year 1: 1.1 million

- Year 2: 860,000
- Year 3: 700,000
- Year 4: 720,000
- Year 5: 820,000

Assuming 9% cost of capital, calculate the Profitability Index.

Question 15:

FutureTech Innovations is a company specialising in cutting-edge technology solutions. The company can only raise \$1 million of finance to invest at T0. The management team has identified five potential projects, each with its own initial investment requirement, NPV, IRR, and discounted payback period. The company's cost of capital is 10%. Below are the details of the projects:

Project	Initial Investment (\$ 000)	NPV (\$ 000)	IRR (%)	Discounted Payback Period (years)
A	300	50	14	4.5
B	200	40	15	3.8
C	500	70	12	5.2
D	400	60	13	4.9
E	250	45	16	3.5